

Database Design & Schema Document

1. Database Architecture

The application utilizes SQLite3 for structured data storage. The database file is located at '05_Project_Development/database/hdi_history.db' and manages historical simulator predictions.

The database is designed with a single high-performance history table to record past scenarios for user comparisons.

2. Entity Schema Details (Table: prediction_history)

- id: INTEGER (PRIMARY KEY, AUTOINCREMENT) - Unique identifier.
- timestamp: DATETIME (DEFAULT CURRENT_TIMESTAMP) - Date/Time of run.
- country: VARCHAR(100) - Name of the country profile.
- life_expectancy: REAL - Health indicator value.
- expected_schooling: REAL - Expected years of schooling.
- mean_schooling: REAL - Average completed education years.
- gni_per_capita: REAL - Gross National Income per capita in USD.
- hdi_score: REAL - Combined predictive score output.
- predicted_category: VARCHAR(50) - Low, Medium, High, or Very High.
- prediction_confidence: REAL - Classifier probability output (%).